STANDATA variance 23-BCV-008(REV1)

Building

Residential exhaust outlet clearance from air intakes

Date Issued: October 2024

Purpose

Provides an alternative solution to Sentence 9.32.3.13.(4) for the minimum clearance requirement for domestic exhaust outlets that discharge air containing moisture, such as kitchen, bathroom, and clothes dryer exhaust outlets, from air intakes.

Discussion

The National Building Code of Canada 2020 introduced changes to Article 9.32.3.13. for minimum clearance requirements related to exhaust vents, soffits, and air intakes. The National Building Code – 2023 Alberta Edition (NBC(AE)) subsequently harmonized with these changes.

The minimum clearance required between a gas vent or oil fill pipe and a mechanically vented air intake was increased from 900 mm to 1 800 mm in Sentence 9.32.3.13.(3).

Additionally, new Sentences (4) to (6) were introduced to address the minimum clearance required from exhaust outlets discharging moisture-laden air to soffits and air intakes. While the national code change rationale for Sentences (4) to (6) specifically explained the application of the 1 800 mm minimum clearance requirement to soffits, it did not address the application to air intakes as provided in Sentence (4). A required 1 800 mm minimum clearance between moist-air exhaust outlets and air intakes for domestic applications was not explained and other standards suggest it can be potentially reduced.

CSA-B149.1, “Natural gas and propane installation code,” (CSA B149.1) provides minimum clearances between clothes dryer vent terminations and air intakes: 900 mm for domestic clothes dryers and 1 800 mm for commercial clothes dryers. As domestic clothes dryer exhaust air typically contains more moisture than bathroom or kitchen exhaust air, this supports the basis for a variance that reduces the 1 800 mm minimum required clearance between a moist-air exhaust outlet and air intake specified by Sentence 9.32.3.13.(4) of the NBC(AE).

As air intakes located above moist-air exhaust outlets are more susceptible to the effects of infiltration (due to the rising of warm, moist air), the reduction in the required minimum clearance from 1 800 mm to 900 mm provided in this variance is only applicable to where the moist-air exhaust outlet is located horizontally to the air intake or above it. This maintains the rationale for the changes to Article 9.32.3.13. in the NBC(AE) while providing appropriate reductions in the required minimum clearance for certain dimensions (see Figure A).

Air intakes specified by Sentence (4) refer to intakes supplying ventilation air to the building and includes those connected to HRV’s/ERV’s or the return air plenum of a furnace. Note that in cases where the fresh air intake connected to the return air plenum of a furnace is less than 8 inches in diameter, it can be considered as a “building opening” rather than an air intake (as the CSA B149.1 standard also allows for).

Moist-air exhaust outlets specified by Sentence (4) include not only bathroom ventilation and clothes dryer exhaust outlets, but kitchen exhaust outlets as well, despite kitchen exhaust outlets not being explicitly mentioned in the Sentence itself or in the rationale for the code change.

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It is important to note that where the minimum clearance requirements contained within a manufacturer's installation instructions differ from the Code or this STANDATA, the more stringent requirements apply.

Code References

Sentences 9.32.3.13.(3) to (6)

9.32.3.13. Outdoor Intake and Exhaust Openings

3) The distance separating air intakes for mechanical ventilation from exhaust outlets that are potential sources of contaminants, such as *gas vents* or oil fill pipes, shall be not less than 1 800 mm.

4) Except as provided in Sentences (5) and (6), exhaust outlets that discharge air containing moisture, such as bathroom ventilation and clothes dryer exhaust outlets, shall be located at least 1 800 mm from air intakes and vented soffits.

5) Where an exhaust outlet referred to in Sentence (4) is located within a soffit, the soffit shall either be unvented, or if vented, the full depth of the soffit shall be blocked for a distance of 1 800 mm on each side of the exhaust outlet.

6) Where an exhaust outlet referred to in Sentence (4) is located in a sidewall less than 1 800 mm from a soffit, a section of the soffit above the exhaust outlet shall be unvented, or if vented, the full depth of the soffit shall be blocked in accordance with the widths stipulated in Table 9.32.3.13.-A, centred over the location of the outlet.

CSA-B149.1 Clauses 7.5.1 and 7.5.2

7.5 Domestic-type clothes dryers

7.5.1
A dryer shall be equipped with a moisture-exhaust duct that terminates outside the ***building***, and the duct shall be constructed of a material that is
(a) ***noncombustible***; or
(b) ***certified*** as meeting the requirements for Class 1 air ducts contained in CAN/ULC-S110.

7.5.2
A moisture-exhaust duct shall not terminate within 3 ft (1 m) in any direction of any ***pressure regulator vent*** termination or fresh-air intake.

Rationale for Clearance Change

Problem

Exhaust from clothes dryers and bathroom fans is allowed to discharge from vents installed in the soffits and in the side walls below the soffits. This warm, moist air can enter the attic space through any vented soffits in the immediate vicinity of the exhaust vent, potentially causing mold, mildew, rot, loss of the insulation's R-value, ice damming, and water infiltration. Approximately five per cent of houses with soffit exhaust vents conforming to the current code requirements show signs of mold, mildew, rot, and water infiltration.

Rationale

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Installing unvented soffits or blocking a section of vented soffits for a certain width on each side of exhaust vents will prevent the infiltration of warm, moist air from the exhaust vents into the attic space, which can cause mold, mildew, rot, loss of the insulation's R-value, ice damming, and water infiltration.

Air containing moisture rises from the exhaust outlets and infiltrates the attic through the soffits, then condensation builds up in the attic over time, resulting in the growth of mold, which causes the deterioration of the ceiling below the roof. Most homeowners do not realize that the roof sheathing on their house is rotting until they replace the roof or notice mold on an interior finish—mold that could be affecting the health of the occupants.

This proposed change is based on the provisions of CSA B139, "Installation code for Oil-Burning equipment" which stipulates that side wall vents need to be a minimum 1.8m from soffits or unvented soffits must be installed.

Application

This variance applies to exhaust outlets that discharge air containing moisture, such as kitchen, bathroom and clothes dryer exhaust outlets, serving an individual dwelling unit or a house with a secondary suite.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

1. The minimum clearances separating exhaust outlets that discharge air containing moisture from air intakes are permitted to comply with Figure A.

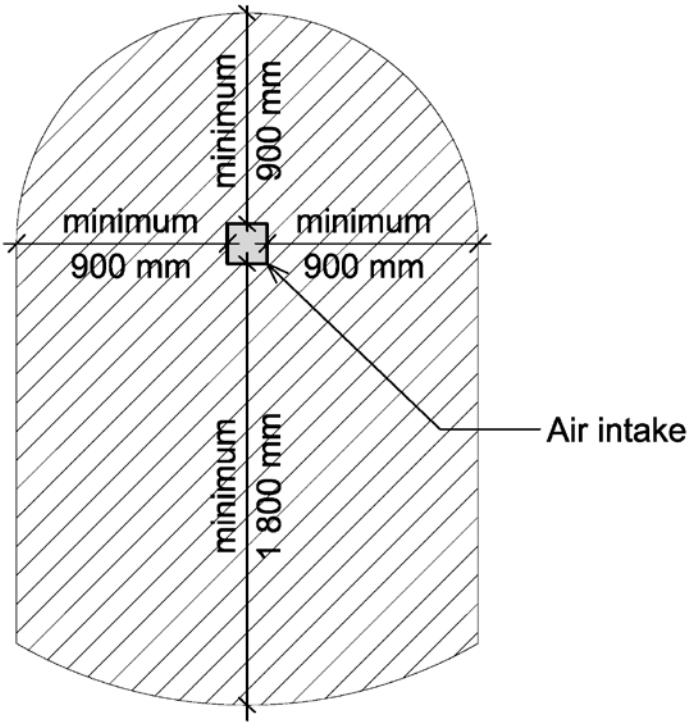


Figure A

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This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

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